



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
25-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

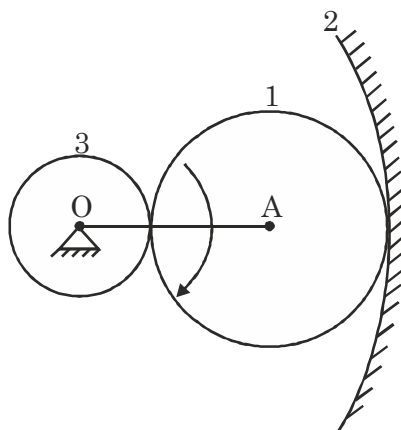
PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Crank OA rotating around point O, leads to movement of wheel 1 of radius $R = 20$ cm rolling inside circle surface 2. Wheel 1, in contact with wheel 3, makes it spin around points O (Fig.).

Find the ratio $\frac{\omega_3}{\omega}$ of angular speed of wheel 3 and that of crank OA if the radius of wheel 3 is $r = 10$ cm.



(A) 1

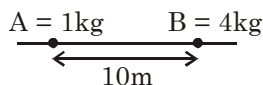
(B) 2

(C) 3

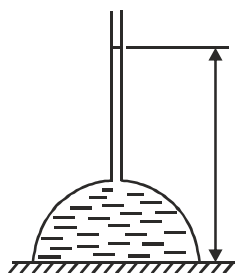
(D) 6

Space for Rough Work

2. Two small bodies of masses 1 kg and 4 kg are at rest on a smooth horizontal surface. When they are at a separation of 10 m, they are released to move under mutual attractive interaction. Then, which of the following statements is **CORRECT**?



- (A) They will never meet
 (B) They will meet at a distance of 6 m from A.
 (C) To an observer at the centre of mass, the particles have equal and opposite momenta.
 (D) They will meet at a distance of 8 m from B.
3. A hollow thin hemisphere of mass M and radius R lies on a flat horizontal rubber surface. In the upper hemisphere there is a small hole in which a long thin tube is inserted (Fig.). Hemisphere tube is filled with water. To what maximum height can be poured the water into the tube?



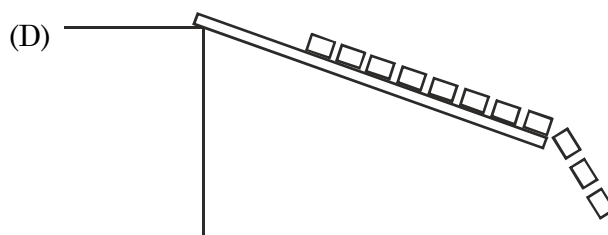
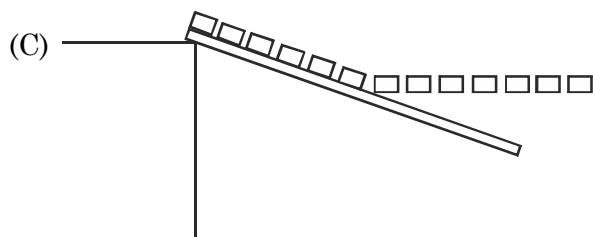
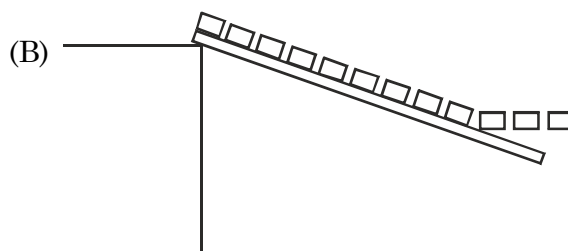
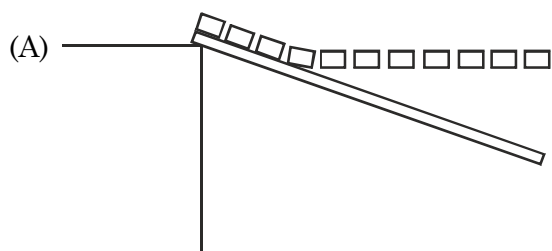
- (A) $\frac{M}{\pi R^2 \rho} + \frac{2}{3}R$ (B) $\frac{M}{\pi R^2 \rho} + \frac{1}{3}R$ (C) $\frac{M}{2\pi R^2 \rho} + \frac{2}{3}R$ (D) $\frac{M}{\pi R^2 \rho} + \frac{4}{3}R$
4. A particle of mass m is moving in potential energy $U = \frac{1}{2}kx^2 + \frac{\alpha}{2x^2}$ where k and α are positive constants. The angular frequency of small oscillations for the simple harmonic motion of the particle about a stable equilibrium position is
- (A) $\sqrt{\frac{k}{m}}$ (B) $\sqrt{\frac{2k}{m}}$ (C) $2\sqrt{\frac{k}{m}}$ (D) $4\sqrt{\frac{k}{m}}$

Space for Rough Work

5. A uniform rigid meter-scale is held horizontally with one of its end at the edge of a table and the other supported by hand. Some coins of negligible mass are kept on the meter scale as shown in the figure.



As the hand supporting the scale is removed, the scale starts rotating about its edge on the table and the coins start moving. If a photograph of the rotating scale is taken soon after, it will look closest to:



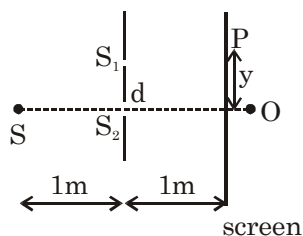
6. There are three planets in circular orbits around a fixed star at distances a , $4a$ and $9a$, respectively in same sense (say anticlockwise). At time $t = 0$, the star and the three planets are in a straight line. The period of revolution of the closest planet is T . What minimum time after $t = 0$ will they again be in the same straight line?
- (A) $8 T$ (B) $27 T$ (C) $108 T$ (D) $512 T$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

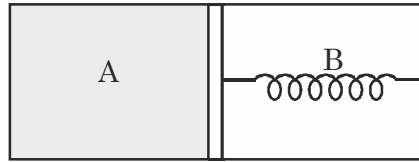
7. In an arrangement shown in the figure, $d \ll D$, where d is the distance of separation of the slits S_1 and S_2 and D is the distance between the slits and the screen. Source S is emitting monochromatic light of wavelength λ . Then



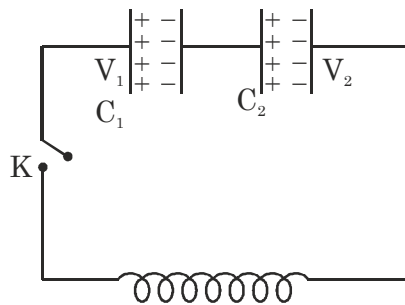
- (A) minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{2}}$.
- (B) minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{3}}$.
- (C) for this value of d , the fringe width is $\sqrt{2\lambda}$.
- (D) for this value of d , the fringe width is $\sqrt{\frac{\lambda}{2}}$.

Space for Rough Work

8. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless insulating piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant K is connected with the piston and the wall of the container as shown. Initially the spring is unstretched. The ideal gas in chamber A is allowed to expand. Let in equilibrium the spring be compressed by a length x_0 , then



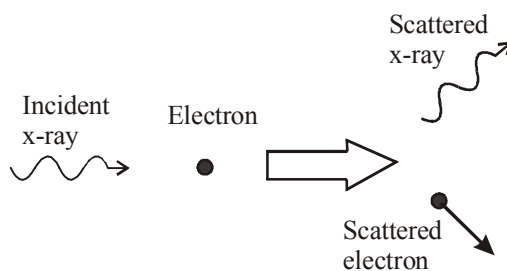
- (A) Final pressure of the gas is Kx_0/S
 (B) Work done by the gas is $\frac{1}{2} Kx_0^2$
 (C) Magnitude of change in internal energy of the gas is $\frac{1}{2} Kx_0^2$
 (D) Temperature of the gas in equilibrium is less than initial temperature
9. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance L as shown in the figure. Initially key k is open. Now key k is closed and current in the circuit starts increasing. When current in the circuit is maximum (Given that C_1V_1 is not equals C_2V_2)



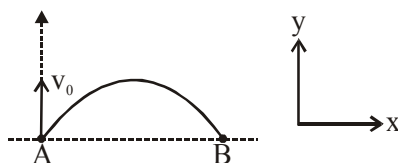
- (A) charge on both the capacitors will be different.
 (B) induced emf in the inductor is zero
 (C) magnitude of potential difference across both the capacitors is same
 (D) energy stored in the inductor is $\frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 + V_2)^2$

Space for Rough Work

10. Further support for the photon model of electromagnetic waves comes from *Compton scattering*, in which x-rays scatter from electrons, changing direction and frequency in the process. Classical electromagnetic wave theory cannot explain the change in frequency of the x rays on scattering, but the photon model can. Suppose an x-ray photon is moving to the right. It has a collision with a slow-moving electron, as in figure. The photon transfers energy and momentum to the electron, which recoils at a high speed. The x-ray photon loses energy, and the photon energy formula $E = hf$ tells us that its frequency must decrease. The collision looks very much like the collision between two particles.



- (A) When the x-ray photon scatters from the electron, its speed stays the same.
 (B) When the x-ray photon scatters from the electron, its speed decreases.
 (C) When the electron is struck by the x-ray photon, its de Broglie wavelength decreases.
 (D) When the electron is struck by the x-ray photon, its de Broglie wavelength stays the same.
11. An electron at point A in the Figure has a velocity $v_0 = 10^7 \text{ ms}^{-1}$. Then (if $e = 1.6 \times 10^{-19} \text{ C}$, $AB = 10 \text{ cm}$ and $m = 9.1 \times 10^{-31} \text{ kg}$)



- (A) magnitude of magnetic induction that will cause the electron to follow the semicircular path from A to B is 1.125 mT
 (B) time taken to go from A to B is $1.57 \times 10^{-7} \text{ s}$
 (C) angular momentum of the electron about A when $t = 0.785 \times 10^{-7} \text{ s}$ is $4.5 \times 10^{-24} \text{ kgm}^2\text{s}^{-1}$.
 (D) angle made by the velocity vector with the x-axis at $t = \frac{3}{8}T$ is $-\frac{\pi}{4} \text{ rad}$.
12. Two identical, stretched wires are vibrated together. They produce few beats per second. T_1 and T_2 denote the tensions in the two wires at any instant and $T_1 > T_2$ (initially). How the change in T_1 or T_2 may be performed so that beat frequency increases.
- (A) T_1 decreased (B) T_1 increased (C) T_2 decreased (D) T_2 increased

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

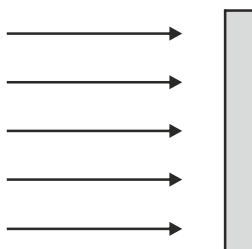
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A smooth insulating fixed guide ring of radius R is in horizontal plane. There is a small bead of mass m and carrying charge q and constraint to move along guide. The guide is placed in a coaxially cylindrically symmetric uniform magnetic field of strength B which is perpendicular to the guide plane. Starting from $t = 0$, the magnetic field is changed to $B = B_0 + \alpha t$. If the force on the bead by the guide at time t is $\frac{\alpha q^2 R t}{X m} (Y B_0 + \alpha t)$, find $\frac{XY}{X + Y}$. Assume gravity to be absent.

Space for Rough Work

2. The shaded area in the figure is the side view of a disk shaped magnetic field region of radius R and thickness d . A parallel electron beam being accelerated by a voltage V is normally incident onto the region. If the spatial distribution of the magnetic field in the region such that the beam is focused to a point along the disk central axis at a distance L ($\gg R$) from the

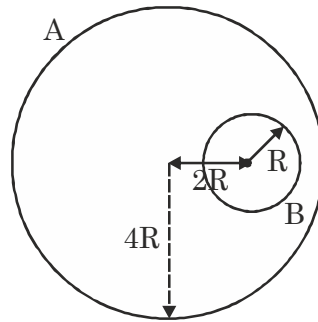
disk is $B = \frac{y^\eta}{d^\alpha L^\beta} \sqrt{\frac{\lambda m V}{e}}$, find the value of $\frac{5\alpha}{\lambda + \beta + \eta}$. Given m and e are the electron's mass and charge and y is the distance of incident electron from axis of disc.



Space for Rough Work

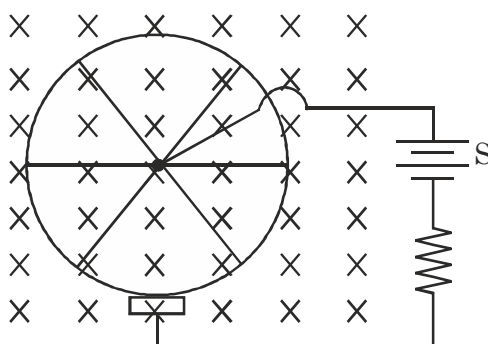
3. Figure shows a uniformly charged spherical shell B of charge ($q_B = q$) kept inside uniformly charged spherical shell A of charge ($q_A = q$). Let electric field due to A and B at any point are

$\vec{E}_A$ and $\vec{E}_B$ respectively. If dV represents elementary volume, then the value of $\frac{\int_{\text{entire space}} \vec{E}_B \cdot \vec{E}_B dV}{\int_{\text{entire space}} \vec{E}_A \cdot \vec{E}_B dV}$ will be



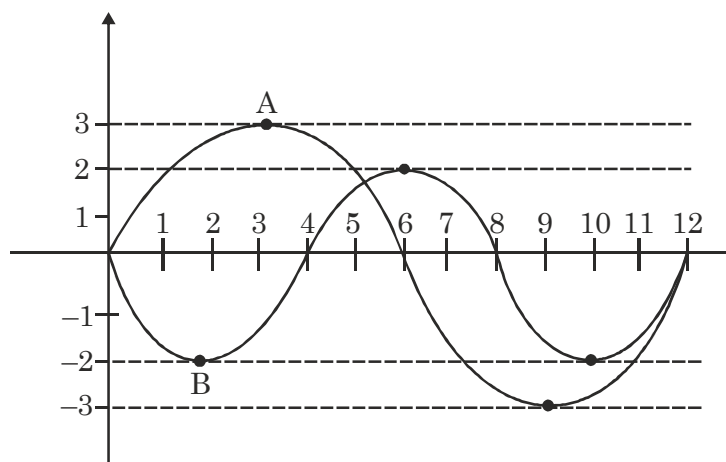
Space for Rough Work

4. The zero of main scale matches with zero of vernier scale. 9th division of main scale matches with 10th division of vernier. No other match is present. After sliding, zero of vernier is found to be in between 3 & 4th divisions of main scale & 5th division of vernier matches with xth division of main scale. The value of $x/3$ is
5. A wheel with six spokes is positioned perpendicular to a uniform magnetic field B of magnitude 0.5 T. The field is directed into the plane of the paper and is present over the entire region of the wheel as shown in the figure. When the switch S is closed, there is an initial current of 6A between the axle and the rim; and the wheel begins to rotate. The resistance of the spokes and of the rim is negligible. The radius of the wheel is 0.2 m. Calculate the initial torque (in Nm) on the wheel.



Space for Rough Work

6. The displacement vs time graph for two sound waves A and B which travel along the same medium are shown in the figure. Their loudness are β_A and β_B . If $X = (\beta_A - \beta_B)$, find the value of X.



7. An ammeter and a voltmeter are connected in series to a battery with an emf $E = 6.0$ V. When a certain resistance is connected in parallel with the voltmeter, the reading of the voltmeter decreases two times, whereas the reading of the ammeter increases the same number of times. Find the voltmeter reading (in volt) after the connection of the resistance.
8. The image of an object formed by a convex spherical mirror is magnified one third in comparison with the object. If the mirror is displaced by 40 cm along its principal optical axis in the direction of light, the image is again found one-third in comparison with the object. Find the radius of curvature of the mirror (in cm).

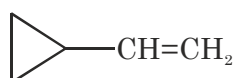
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Consider the following compounds :-



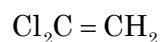
I



II



III



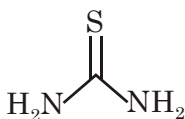
IV

The correct order of reactivity with $\text{Br}_2 + \text{CH}_3\text{COOH}$ is

- (A) $\text{I} < \text{II} < \text{III} < \text{IV}$ (B) $\text{I} > \text{II} > \text{III} > \text{IV}$ (C) $\text{II} > \text{I} > \text{III} > \text{IV}$ (D) $\text{II} > \text{I} > \text{IV} > \text{III}$
2. The compounds those would not respond to tests of both nitrogen and sulfur with sodium fusion extracts are?



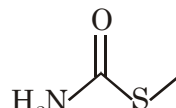
I



II



III



IV

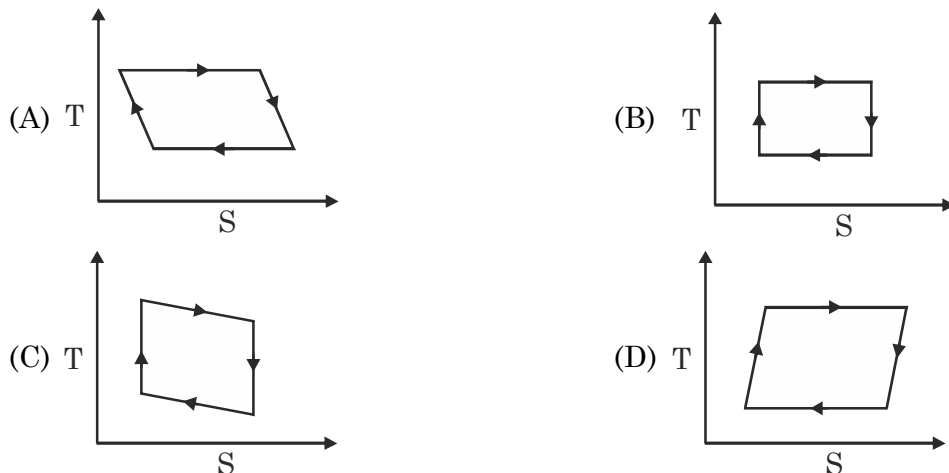
- (A) I and III (B) III and IV (C) I and IV (D) II and IV
3. Which of the following does liberate hydrogen gas when treated with water under suitable condition ?
- (I) B_2H_6 (II) $\text{B}_3\text{N}_3\text{H}_6$ (III) SiH_4 (IV) XeF_4
 (A) I, III, IV only (B) II, III only (C) I, II, IV only (D) I, II, III only

Space for Rough Work

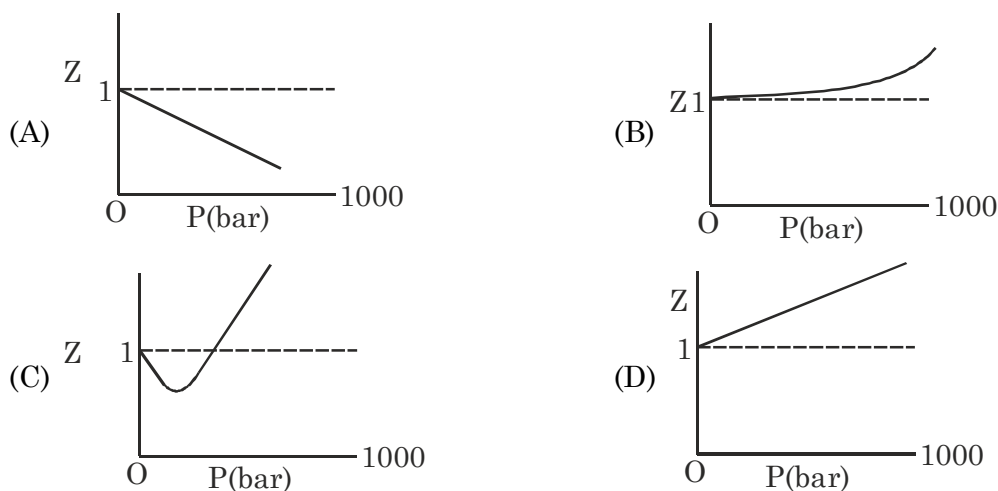
4. Select the correct statement when potassium ferrocyanide is treated with concentrated sulphuric acid ?

(A) Fe^{2+} of $[\text{Fe}(\text{CN})_6]^{4-}$ is oxidised to FeO_2^- (B) CO gas is evolved
(C) HCOOH is one of final product (D) HCN gas is one of the final products

5. The graph that represents the temperature (T) – entropy (S) variation of a carnot cycle is



6. The Boyle temperature (T_B) is defined as the temperature at which the properties of a real gas coincide with those of an ideal gas in the low pressure limit. The graph that shows the pressure dependance of the compressibility factor (Z) for a real gas at T_B is



Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

7. Which of the following compounds exist as polymeric solid at room temperature ?

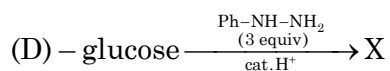
- (A) NaPO_3 (B) H_3BO_3 (C) BeH_2 (D) BeCl_2

8. The reactions which produce $\text{Na}_2\text{S}_2\text{O}_3$ are

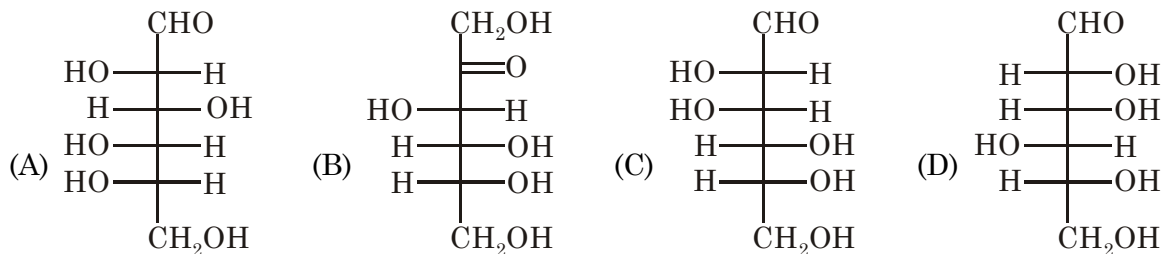
- (A) $\text{Na}_2\text{S} + \text{I}_2 + \text{Na}_2\text{SO}_3 \longrightarrow ?$ (B) $\text{Na}_2\text{S} + \text{Na}_2\text{CO}_3 + \text{SO}_2 \longrightarrow ?$
(C) $\text{S} + \text{NaOH} \longrightarrow ?$ (D) $\text{Na}_2\text{SO}_3 + \text{S} \xrightarrow{\Delta} ?$

Space for Rough Work

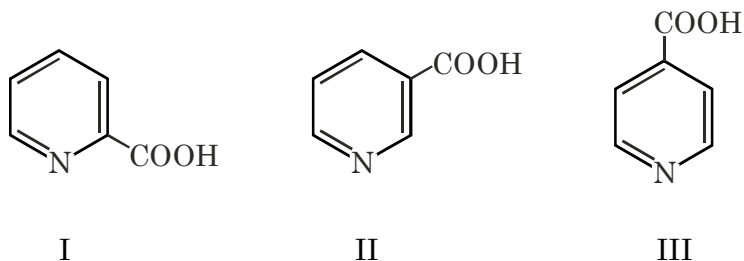
9. Consider the following reaction :



Among the following, the compound(s) whose osazone derivative(s) will have the same melting point as that of X is/are



10. Consider the following compounds



Select the correct statements

- (A) (I) undergoes decarboxylation faster than II or III on heating
- (B) (I) when heated in the presence of acetophenone produces a compound having tertiary alcoholic group
- (C) Compound (I) acts as bidentate ligand
- (D) The compound (III) has the highest melting point.

Space for Rough Work

11. A complex is represented as $\text{CoCl}_3 \cdot x\text{NH}_3$. Its 0.1 molal solution in water shows $\Delta T_f = 0.558^\circ$. If the cryoscopic and ebullioscopic constants for H_2O are $1.86 \text{ K}\cdot\text{kg mol}^{-1}$ and $0.52 \text{ K}\cdot\text{kg mol}^{-1}$ respectively, identify the correct versions assuming 100% ionisation of the complex.
- (A) Hexaaminecobalt(III) chloride is the IUPAC name for the complex
(B) The above solution will boil between 100°C to 101°C
(C) The central metal ion is d^2sp^3 hybridised
(D) The coordination sphere is incapable of exhibiting geometrical isomerism
12. You have a mixture of three amino acids : A($pI = 3$), B($pI = 5$) and C($pI = 10$), where pI represents their isoelectric points. Under electrophoresis at $pH = 8$, the movement of each component of the mixture is considered. Which of the following options are incorrect regarding movement acids under electrophoresis :
- (A) A and B to anode; C to cathode
(B) A to anode; B stationary; C to cathode
(C) A to cathode ; B stationary; C to anode
(D) A to anode; B and C to cathode

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. For 3s orbital of hydrogen atom, the normalised wave function is given by

$$\psi_{3s} = \frac{1}{81\sqrt{3\pi}} \left(\frac{1}{a_0} \right)^{3/2} \left[27 - \frac{18r}{a_0} + \frac{2r^2}{a_0^2} \right] e^{-\frac{r}{3a_0}}$$

The above mentioned orbital (3s) has two nodes at $2a_0$ and xa_0 . Find the value of $x / 4$.

[Use : $3\sqrt{3}=5$]

2. The emf of the cell,

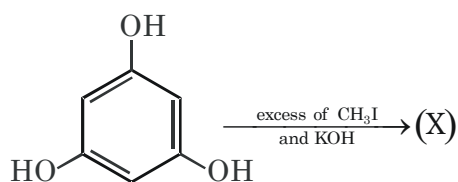
$M(s) | M_nX_n(0.01M) || M_nX_n(0.1M) / M(s)$ is found to be 0.0295 V at 25°C. If M_nX_n is a soluble salt, find the value of $2n / 3$.

Space for Rough Work

3. 2.616 gm of an element 'X' is heated with NaOH and NaNO_3 to produce Na_2XO_2 and NH_3 . Ammonia produced is absorbed in 100ml of 0.11 M H_2SO_4 and excess of acid is back titrated with 48 ml of 0.25 M NaOH. If N moles of hydrogen gas would be liberated when 2 moles of 'X' is treated with excess of NaOH, find the value of $\frac{1}{2N}$.
- [Atomic mass : Cr = 52, Zn = 65.4, Fe = 56, Cu = 63.5, Ni = 28]
4. When aqueous solution of sodium phthalate is electrolysed the product (hydrocarbon) at anode has x degree of unsaturation. Find the value of $\frac{x}{2}$.

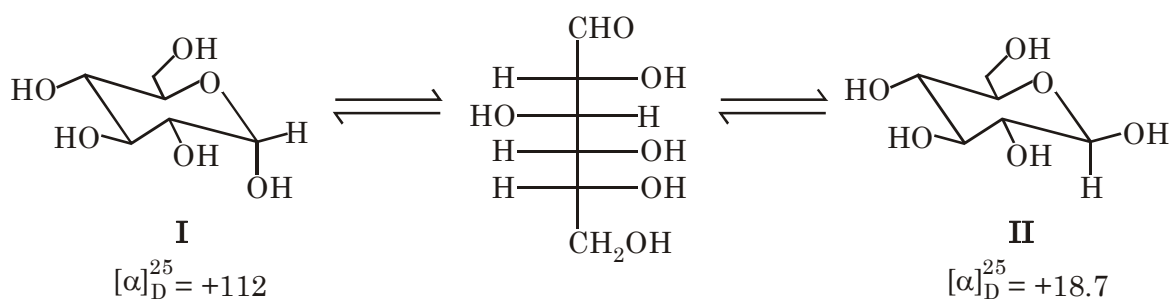
Space for Rough Work

5. Consider the following reaction given below



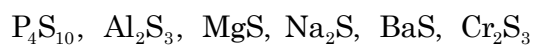
Find the value of $\frac{n}{4}$ where n is the total number of methyl groups present in one molecule of (X).

6. Specific rotations of freshly prepared aqueous solutions of **I** and **II** are +112 and +18, respectively. On standing the optical rotation of aqueous solution of **I** slowly decreases give a final value of +52.7 due to equilibration with **II**. Under this state of equilibrium, what the ratio **II** : **I**?



Space for Rough Work

7. Find the value of $\frac{n}{5}$ where n is the total number of compounds among the following which produce H_2S gas on treatment with water.



8. Consider the following compounds / ions SnCl_2 , HgCl_2 , H_2O , NOCl , H_2S , I_3^+ , OF_2 , NH_2^- , C_2H_2

Find the value of $\frac{x}{2}$ where x is the total number of species having bent shape around central atom.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A variable plane cuts off intercepts from the co-ordinate axes which are equal to the roots of the equation $x^3 + 5x = p - qx^2$ (p, q are real numbers). The locus of the foot of the perpendicular from the origin to the plane is
 (A) $(x^2 + y^2 + z^2)^2 (xy + yz + zx) = 5$ (B) $(x^2 + y^2 + z^2)^4 \left(\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} \right) = 5$
 (C) $(x^2 + y^2 + z^2)^2 \left(\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} \right) = 5$ (D) $(x^2 + y^2 + z^2)(xy + yz + zx) = 5$
2. Consider $f(x) = x^4 - 3x^3 + 2$ and let $h(x) = \frac{f(x)}{g(x)}$ be a rational function such that
 (i) Domain of $h(x)$ is $x \in \mathbb{R} - \{-1, 1\}$.
 (ii) $g(x)$ does not have any repeated linear factors.
 (iii) As x approaches infinity, $h(x)$ also approaches $-\infty$.
 (iv) $\lim_{x \rightarrow 1} h(x) = \frac{1}{2}$ then $\lim_{x \rightarrow 0} \frac{g(\cos x)}{5 \ln(\sec^2 x)}$ is equal to -
 (A) 1 (B) 2 (C) 3 (D) 5

Space for Rough Work

3. If A is a square matrix of order 5 and $9A^{-1} = 4A^T$ then number of digit before decimal in $|\text{adj}(\text{adj}(\text{adj}A))|$ (where A^{-1} , A^T and $\text{adj}(A)$ denotes the inverse, transpose and adjoint of matrix A respectively) is : ($\log 3 = 0.477$, $\log 2 = 0.303$)
 (A) 56 digits (B) 60 digits (C) 58 digits (D) 53 digits
4. Let ' d ' be defined as $d = \sqrt{a^2 + 5a + 8} - \sqrt{a^2 - 3a + 4}$, $a \in R$. The number of integral values of d is
 (A) 7 (B) 9 (C) 13 (D) 15
5. In ΔABC with usual notation, if a, c, A are given and $b_1 = 2b_2$ where b_1, b_2 are two values of the third side then the value of $\frac{c\sqrt{1+8\sin^2 A}}{a}$ is less than
 (A) 1 (B) 2 (C) 3 (D) 4
6. Sum of the solutions of the equation $\sin^2\theta \tan\theta + \cos^2\theta \cot\theta - \sin 2\theta = 1 + \tan\theta + \cot\theta$, where $\theta \in (0, \pi) - \left\{\frac{\pi}{2}\right\}$, is -
 (A) 3π (B) $\frac{3\pi}{2}$ (C) $\frac{5\pi}{2}$ (D) 7π

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

7. If z is a unimodular complex number such that

$$\operatorname{Re}(z-1) + \operatorname{Re}(z^2) = \int_0^{\pi/2} \sin x \ln |\sin x - \cos x| dx, \text{ then } z \text{ can satisfy}$$

- (A) $z + \bar{z} = -2$ (B) $z + \bar{z} = 1$ (C) $\arg(z) = \frac{\pi}{3}$ (D) $\arg z = \pi$

Space for Rough Work

8. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. If $f(x)$ is differentiable on $(0, 1)$ & $\int_0^1 f(x) dx = 0$, then which option(s) is/are correct :-

(A) $\int_0^c f(x) dx = f(c)$ for at least one c in $(0, 1)$

(B) $\int_0^c f(x) dx = f(c)$ for all $c \in (0, 1)$

(C) $\int_0^c f(x) dx = -\left(\frac{1-c}{c}\right)f(c)$ for at least one c in $(0, 1)$

(D) $\int_0^c f(x) dx = -\left(\frac{1-c}{c}\right)f(c)$ for all $c \in (0, 1)$

9. Consider a curve $C : y = \sqrt{\frac{3x^2 - 21}{7}}$ and a line L of slope 1, which is tangent to C at point A , then

(A) Equation of L is $y = x + 2$

(B) Equation of L is $y = x - 2$

(C) Equation of circle passing through origin and touches the curve C at A is $x^2 + y^2 + x - 41y = 0$

(D) Equation of circle passing through origin and touches the curve C at A is $4x^2 + 4y^2 + x - 41y = 0$

Space for Rough Work

10. Let $|a|, |b| < 1$, where a and b are complex number, then

- (A) $|a - b| < |1 - \bar{a}b|$ (B) $|a - b| > |1 - \bar{a}b|$ (C) $\left| \frac{a - b}{1 - \bar{a}b} \right| < 1$ (D) $\left| \frac{a - b}{1 - \bar{a}b} \right| > 1$

11. Let $f(x) = \frac{x^3 - 4}{(x - 1)^3}$ then

(A) $\lim_{x \rightarrow -\infty} \sqrt{f(x)} = 1$

(B) $x = -2$ is the point of local minimum.

(C) the equation $f(x) = c$ will have three real and distinct roots for $c \in \left(\frac{4}{9}, 1\right) \cup (1, 4)$.

(D) the line $y = 1$ intersects the graph of $f(x)$ at exactly one point.

12. Given a differentiable function $f(x)$ satisfying the integral equation

$$f(x) = \int_0^x e^{x-y} f'(y) dy - (x^2 - x + 1)e^x \forall x \in \mathbb{R}, \text{ then-}$$

(A) $f(x)$ is local minimum at $x = -\frac{1}{2}$

(B) $f(x)$ has exactly one asymptote

(C) If $f(x)$ has an extremum point at $x = x_1$ and point of inflection at $x = x_2$, then $|x_1 - x_2| = 1$

(D) $|f'(x) - f(x)| = |f(x) - f''(x)|$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+										☒										
☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒										
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

										☒	-									
☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒										
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Two circles C_1 and C_2 both passes through the point A(1, 2) and E(2, 1) and touch the line $4x-2y = 9$ at B and D respectively. The possible coordinates of a point C such that the quadrilateral ABCD is a parallelogram is (a, b) then the value of $|ab|$ is :
- Given A(0,0,1), B(2,1,2) be two points in 3-D space also P(α , β) ($\alpha > 0$, $\beta > 0$) be a point on the

xy-plane such that $\angle APB$ is maximum, then $\left[\frac{\alpha + 3\beta}{5} \right] + \left(\frac{\alpha + 2}{\beta + 1} \right)$ is equal to

(where $[.]$ greatest integer function and A,P,B are non collinear points)

Space for Rough Work

3. A pack of playing card is consist of 51 cards. Cards are drawn one by one from the pack without replacement. If first 4 cards drawn consist of only one honour card, then the probability that the missing card being honour card is $\frac{a}{b}$ (where a and b are co-prime and all Jacks, Queens, Kings and Aces are considered as honour card) then $\frac{a}{b}$ is _____ .
4. A certain kind of bacteria either die, split into two or split into three bacteria. All splits are exact copies. The chances of dying is $\frac{1}{4}$, the chances of splitting into two is $\frac{1}{2}$ and splitting into three is $\frac{1}{4}$. If the probability that it survives for infinite length of time is $\frac{m - \sqrt{13}}{n}$ ($m, n \in \mathbb{N}$), then the value of (m/n) is
5. Let $f(x, y)$ is defined for $x, y \in \mathbb{N}$ and satisfy $f(x, x) = x$
- $$f(x, x+y) = \left(1 + \frac{x}{y}\right) \cdot f(x, y)$$
- $$f(x, y) = f(y, x) . \text{ Then } f(52, 14) \text{ is}$$

Space for Rough Work

6. There are five different boxes and seven different balls. The number of ways in which these balls can be distributed so that box 2 and box 4 contains only 1 ball each and at least 1 box is empty is N . (Order of putting the balls in the boxes is NOT considered). Then the value of $N/100$ is _____ .
7. Let $f(x)$ and $g(x)$ are differentiable function of x such that

$$x^2 (gf'(x)) \cdot (f'g(x)) \cdot (g'(x)) = (1 - 2x^2)(g'f(x)) \cdot (f'(x)) f(g(x)) \quad \forall, x \in R.$$
 Moreover $g(x)$ is positive and $f(x)$ is non negative and $\int_0^k f(g(x))dx = 1 - e^{-k^2}$, then the value of $\ln\left(\frac{gf(3)}{gf(2)}\right)$ is _____
8. If the product of all possible values of $c \in \mathbb{N}$ for which $x^2 - 20x + c = 0$ has always a real root, is $n!$ then sum of digits of n , is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–1 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator